Nitesh Saini

ce24btech11045@iith.ac.in | +91 9670225206

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.50
SHRI CHHOTELAL I C KHANPUR INAYATULLAH BARABANKI	Class XII (Board of High School and Intermediate Education, Uttar Pradesh)	2023	88.80%
SH C L I C KHANPUR INAYATULLAH BARABANKI	Class X (Board of High School and Intermediate Education, Uttar Pradesh)	2021	94.17%

SKILLS

Programming languages - C, C++, Python.

Computer Science Basics : DSA (Data Structure Algorithm), **OOPs**(Object-Oriented Programming),**SQL**(Structured Query Language).

Tools - Solid edge, Git, Fusion 360.

Soft Skills - Problem solving, Communication, Critical thinking, Co-ordination, Teamwork, Adaptability.

RELEVANT COURSES

Civil Engineering(IITH Curriculum) - Digital Fabrication, Engineering Mechanics, Strength of Materials, Structural Analysis, Surveying, Fluid Mechanics, Geotechnical Engineering, Hydraulics.

Mathematics (IITH Curriculum) - Calculus, Linear Algebra, Differential Equations, Probability, Transform Techniques, Complex Variables, Statistics.

Computer Science - Introduction to Programming.

PROJECTS

Banking Management System

- Developed a console-based **C++** Banking Management System using **OOP** principles and a modular, multi-file architecture.
- Implemented **secure account authentication** along with core operations like deposits, withdrawals, and fund transfers.
- Designed **persistent account storage** using C++ file handling for automatic data saving and loading across sessions.
- Built a **transaction history system** utilizing dedicated log files for efficient tracking and retrieval of user operations.

Multithreaded HTTP Server | C++, POSIX Sockets, Multithreading

- Built a multithreaded **HTTP/1.1** server from scratch in C++ using POSIX sockets and TCP/IP networking.
- Implemented HTTP request parsing, routing, static file serving, **MIME type detection**, Keep-Alive connections, and custom **403/404 error pages**.
- Developed a thread pool using std::thread, mutexes, and condition variables to handle multiple concurrent client requests efficiently.

- Added query parameter parsing, form data handling, request logging, HTTP redirects, and directory traversal protection.
- Used **CMake** for project build configuration and **Git/GitHub** for version control.

EXTRACURRICULAR

NSO chess player.

Sports (Volleyball) - Regular player; secured Gold Medal in **Institute Volleyball League (IVL)**, demonstrating teamwork and competitive spirit.